Yao Ji

H. Milton Stewart Postdoctoral Fellow

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Last updated: August 2026

Employment

Georgia Institute of Technology, Atlanta, Georgia — Oct. 2024–Present
H. Milton Stewart Postdoctoral Fellow
H. Milton Stewart School of Industrial and Systems Engineering (ISyE)
Mentor: Guanghui (George) Lan

Education

Purdue University, West Lafayette, Indiana — Aug. 2019–Aug. 2024
Ph.D. in Operations Research
Advisor: Harsha Honnappa
Committee: Harsha Honnappa, Raghu Pasupathy, Alex L. Wang, and Gesualdo Scutari
Dissertation: *High-Dimensional Inference over Networks: Statistical and Computational Guarantees*

Beijing Normal University, Beijing, China — Aug. 2016–Jun. 2019
M.S. in Probability and Mathematical Statistics
Advisor: Wenming Hong
Thesis: *Conditional Limit Theorem for Bellman–Harris Branching Process*

Beijing Normal University, Beijing, China — Aug. 2012–Jun. 2016
B.S. in Mathematics
Thesis: *Conceptual New Proofs of Geometric Convergence of Moment Generating Function for Galton–Watson Process in the Noncritical Case*

Research Interests

Parameter-free optimization; variational inequalities; stochastic optimization; gradient minimization; high-order optimization; reinforcement learning; reinforcement learning in health care.

Research Papers

The candidate's name is in bold.

Job-Market Paper

Yao Ji and Guanghui Lan
Stochastic Auto-conditioned Fast Gradient Methods with Optimal Rates
Job-market paper | Journal manuscript | Under review, *Mathematical Programming* — first-round review; arXiv:2604.06525 (2026).

Manuscripts Under Review

Yao Ji and Guanghui Lan
High-order Accumulative Regularization for Gradient Minimization in Convex Programming
Journal manuscript | R&R, *SIAM Journal on Optimization* — revision in progress; arXiv:2511.03723 (2025).

Yao Ji, Guanghui Lan, and Jason Zhu
Complexity of Strong Solutions for Stochastic Variational Inequalities
Journal manuscript | Under review, *Mathematical Programming* — first-round review.

Research Papers

Published Journal Articles

Yao Ji, Gesualdo Scutari, Ying Sun, and Harsha Honnappa
Distributed Sparse Regression via Penalization
Journal article | Published, *Journal of Machine Learning Research*, 24(272):1–62, 2023. Paper.

Yao Ji, Gesualdo Scutari, Ying Sun, and Harsha Honnappa
Distributed (ATC) Gradient Descent for High Dimension Sparse Regression
Journal article | Published, *IEEE Transactions on Information Theory*, 69(8):5253–5276, 2023. Paper.

Vladimir A. Vatutin, Wenming Hong, and **Yao Ji**
Reduced critical Bellman–Harris branching processes for small populations
Journal article | Published, *Discrete Mathematics and Applications*, 28(5):319–330, 2018. Paper.

Preprint

Ruqi Bai, **Yao Ji**, Zeyu Zhou, and David I. Inouye
From Invariant Representations to Invariant Data: Provable Robustness to Spurious Correlations via Noisy Counterfactual Matching
Preprint | arXiv:2505.24843 (2025).

Conference and Workshop Paper

Ruqi Bai, **Yao Ji**, Mingyu Kim, Easton Currie, Zeyu Zhou, and David I. Inouye
Provable Robustness to Spurious Correlations via Invariant Data for Robust Finetuning
Workshop paper | Accepted, AISTATS 2026 Workshop on Causality in the Age of AI Scaling. Paper.

Teaching Experience

Georgia Institute of Technology, ISyE
Instructor

ISyE 4106: Senior Design	Fall 2026

Purdue University, Industrial Engineering
Teaching Assistant

IE 335: Operations Research	Spring and Fall 2023
IE 330: Probability and Statistics in Engineering	Fall 2022
IE 590: Introduction to Optimization Algorithms (graduate-level)	Fall 2022

Beijing Normal University, School of Mathematical Sciences
Co-Lecturer

Large Deviation Theory	Spring 2018
Brownian Motion	Fall 2017
Random Walk in Random Environment	Spring 2017
Galton–Watson Branching Process	Fall 2016

Teaching Assistant; Outstanding Teaching Assistant Award recipient

Measure Theory I	Fall 2017 and Fall 2018
Measure Theory II	Spring 2017 and Spring 2018
Stochastic Calculus for Finance (graduate-level)	Fall 2016

Outreach

Instructor: Pre-College Reinforcement Learning Course for High School Students, Georgia Tech	Summer 2026
Judge: STEM Spark: INFORMS K–12 Poster Competition	2026
Judge: Georgia Tech 2026 CRIDC Poster Competition	2026

Awards and Honors

H. Milton Stewart Postdoctoral Fellowship, Georgia Institute of Technology	2024
Lee A. Chaden Fellow IE Scholarship, Purdue University	2023
Graduate School Summer Research Grant, Purdue University	2023
Travel Grant, School of Industrial Engineering, Purdue University	2022, 2023
Ross Fellowship, Purdue University	2019, 2020
Dr. Theodore J. and Isabel M. Williams Fellowship in Industrial Control Systems, Purdue University	2019
First Prize Scholarship (2nd of 53, School of Mathematics), Beijing Normal University	2018
First Prize Scholarship (1st of 12, Markov Process), Beijing Normal University	2017
Outstanding Teaching Assistant for Measure Theory, Beijing Normal University	2017
Outstanding Undergraduate Thesis, School of Mathematics, Beijing Normal University	2016
First Prize Scholarship (top 5%), School of Mathematics, Beijing Normal University	2015
Second Prize Scholarship, Beijing Normal University	2014
Second Prize, China Undergraduate Mathematical Modeling Contest (top 5%)	2014

Talks

Stochastic Auto-conditioned Fast Gradient Methods with Optimal Rates
- MOPTA, Lehigh University, Bethlehem, PA — Aug. 2026
- Clemson University, School of Mathematical and Statistical Sciences, Clemson, SC — Apr. 2026
- INFORMS Optimization Society Conference, Atlanta, GA — Mar. 2026

High-order Accumulative Regularization for Gradient Minimization in Convex Programming
- INFORMS Annual Meeting, Atlanta, GA — Oct. 2025

Poster Presentations

Stochastic Auto-conditioned Fast Gradient Methods with Optimal Rates
- Southern California OR/OM Day, USC Marshall, Los Angeles, CA — May 2026

Transformer-Based Patient Response Modeling Across All Perioperative Phases for Cardiac Surgery-Associated Acute Kidney Injury Management
- Society of Cardiovascular Anesthesiologists Annual Meeting, Nashville, TN — Apr. 2026

Distributed (ATC) Gradient Descent for High Dimension Sparse Regression
- SIAM Conference on Mathematics of Data Science, Atlanta, GA — Oct. 2024
- International Symposium on Mathematical Programming (ISMP), Montréal, Canada — Jul. 2024
- Midwest Machine Learning Symposium, University of Illinois Chicago, Chicago, IL — May 2023
- Statistics and Optimization in Data Science Workshop, Purdue University, West Lafayette, IN — May 2023

Distributed Sparse Regression via Penalization
- Cornell ORIE Young Researchers Workshop, Ithaca, NY — Oct. 2023
- International Conference on Continuous Optimization (ICCOPT)/MOPTA, Lehigh University, Bethlehem, PA — Jul. 2022

Service

Reviews: Mathematical Programming; *SIAM Journal on Optimization*; *Operations Research*; *Journal of the American Statistical Association*; *Stochastic Systems*; *Journal of Global Optimization*; *Journal of Optimization Theory and Applications*; and *IEEE Transactions on Automatic Control*.

Conference reviews: IEEE International Symposium on Information Theory.

Leadership: President, INFORMS Student Chapter Board, School of Industrial Engineering, Purdue University.